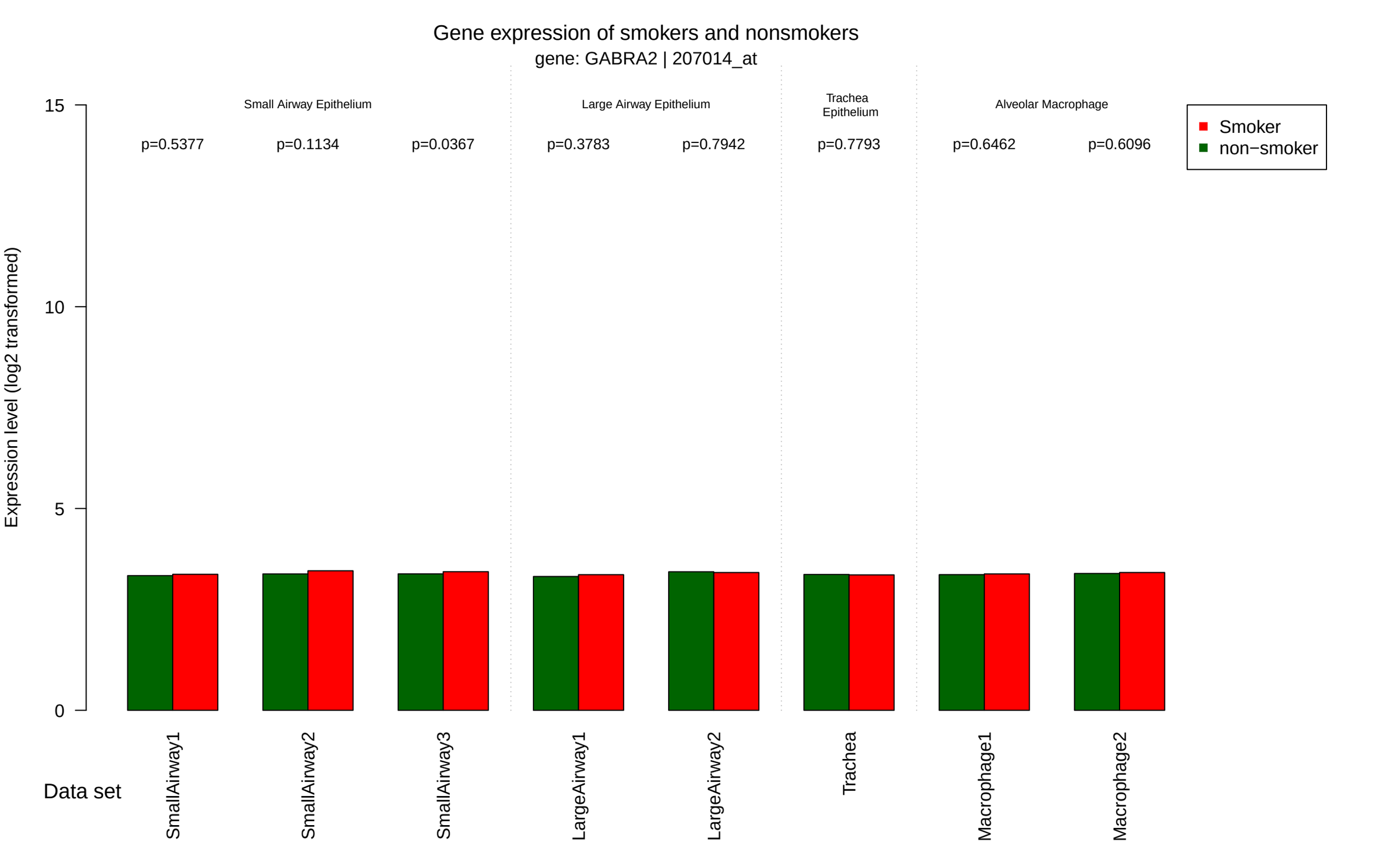
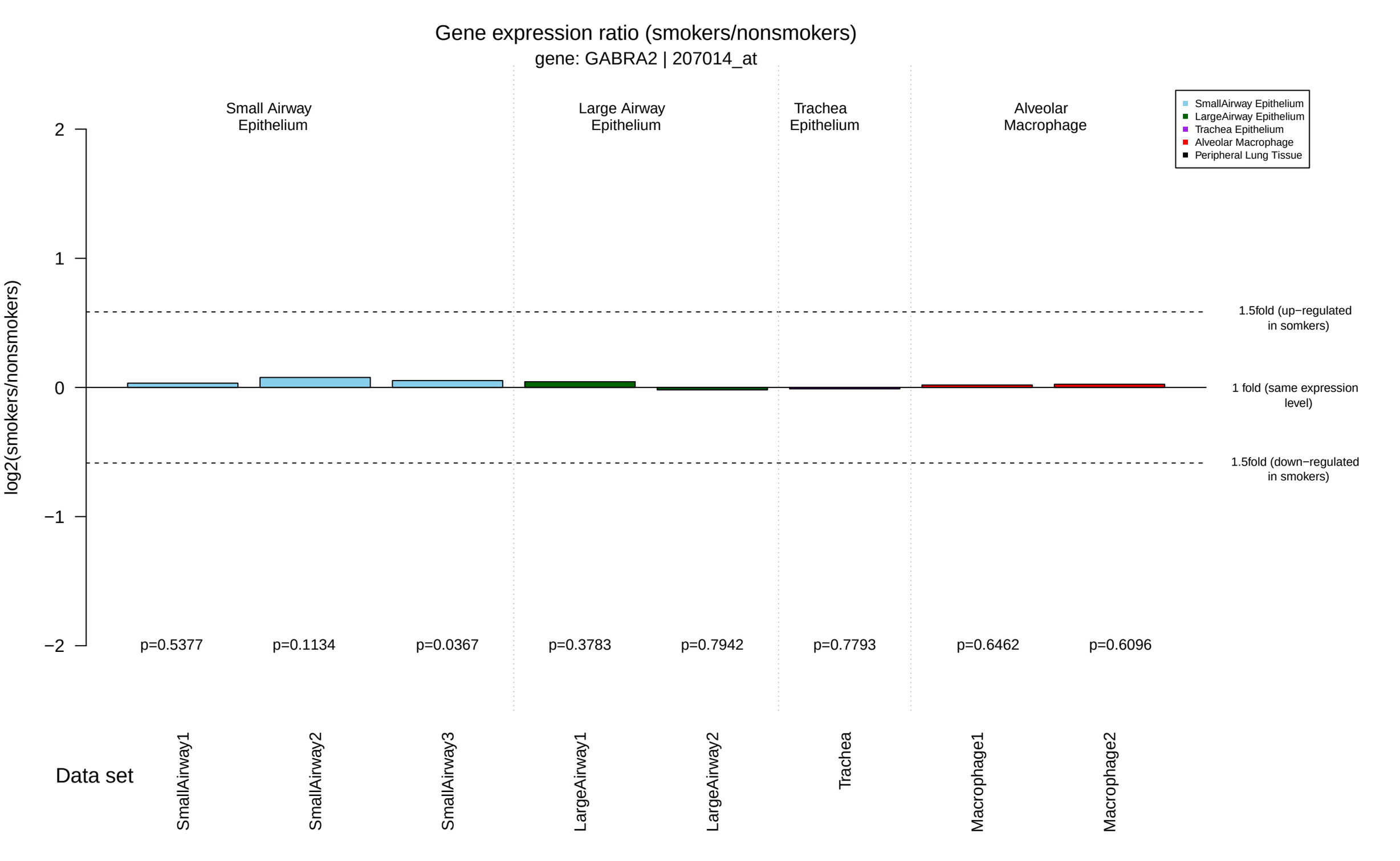
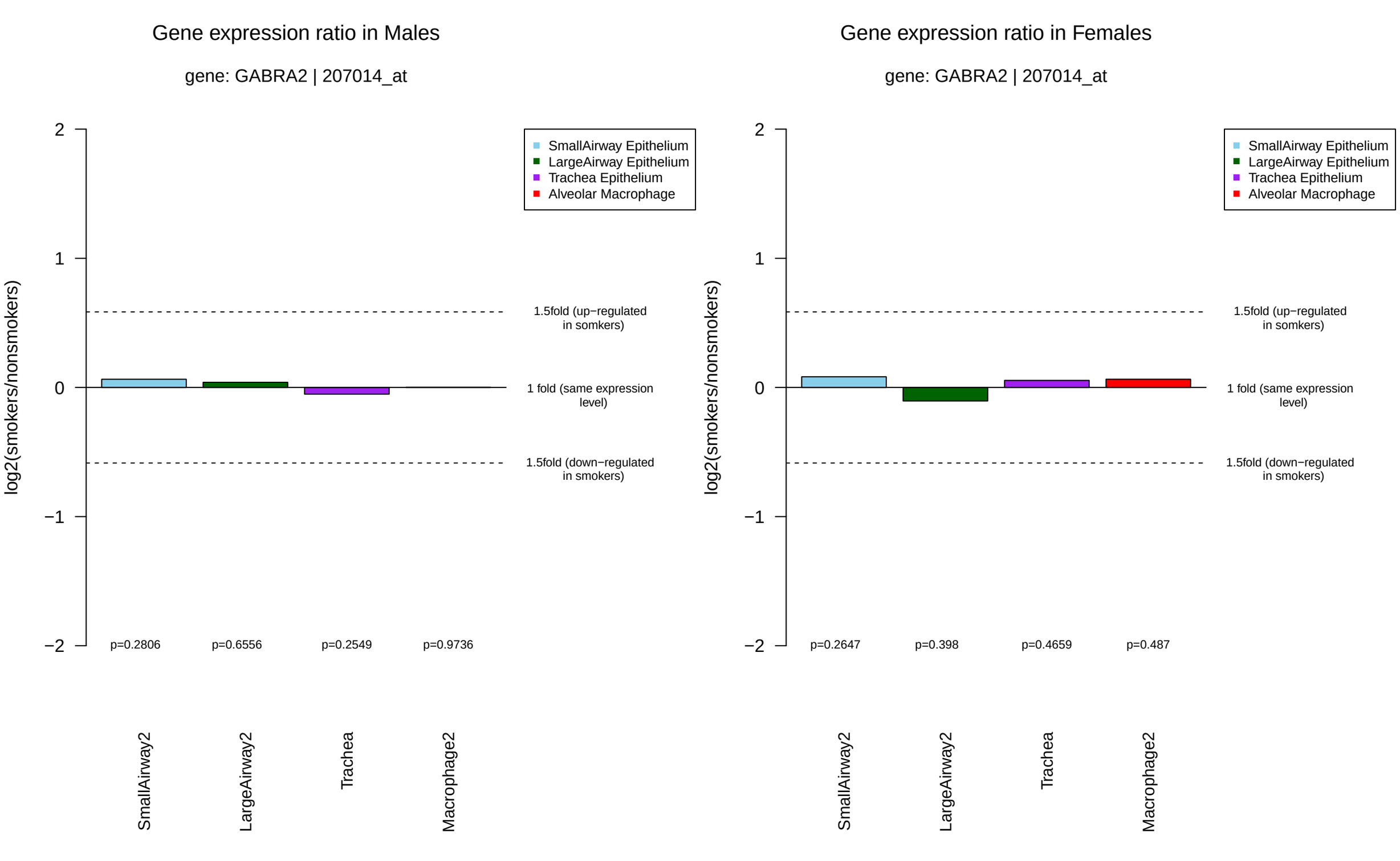
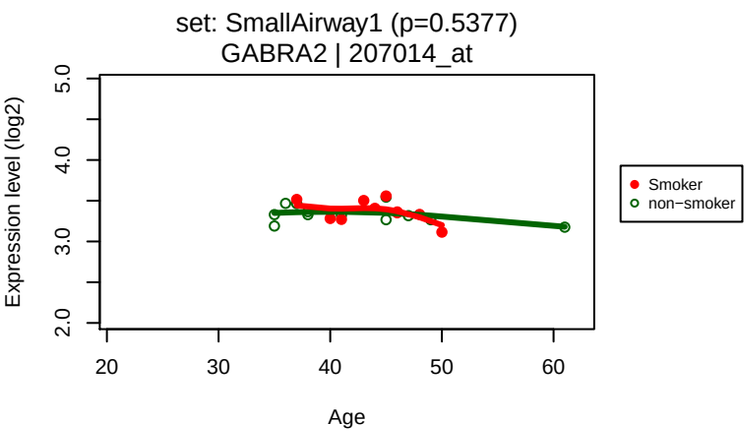
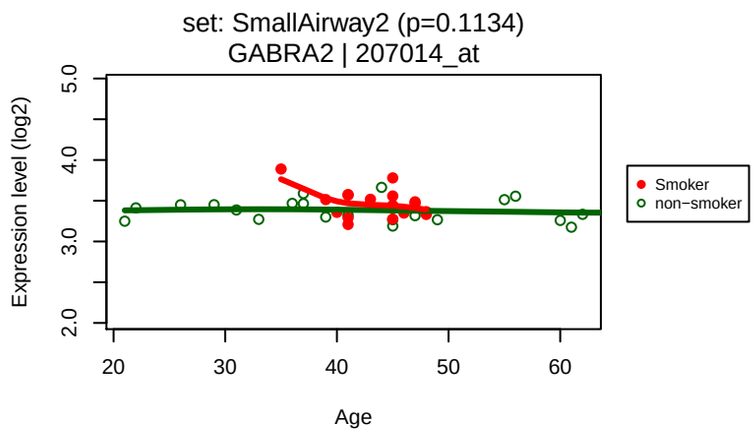
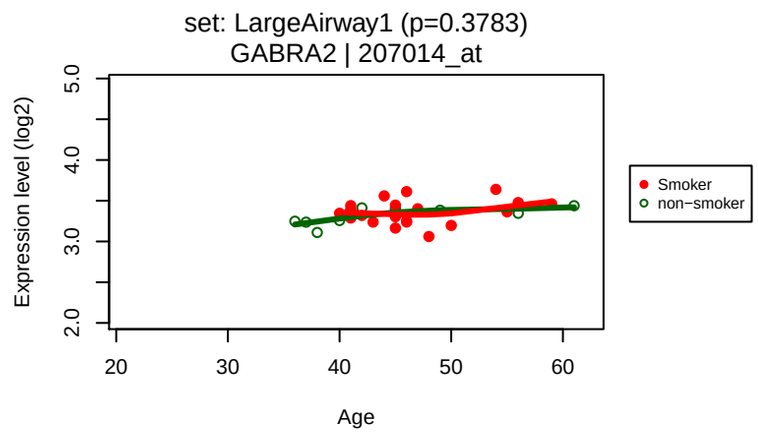

Gene expression of smokers and nonsmokers
gene: GABRA2 | 207014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5377
p=0.1134
p=0.0367
p=0.3783
p=0.7942
p=0.7793
p=0.6462
p=0.6096
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

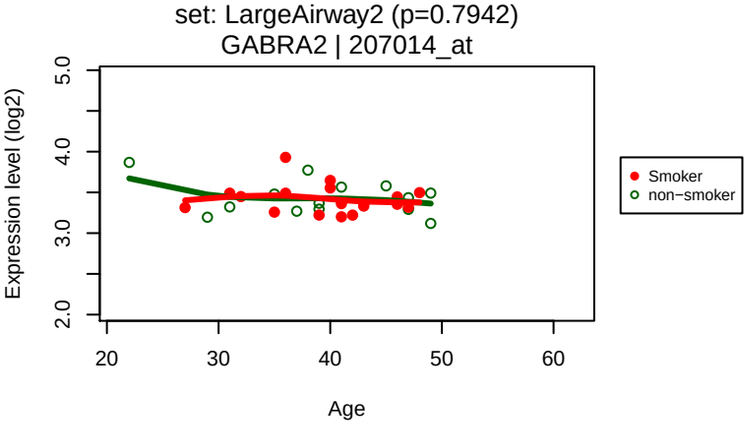

Gene expression ratio (smokers/nonsmokers)
gene: GABRA2 | 207014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5377
p=0.1134
p=0.0367
p=0.3783
p=0.7942
p=0.7793
p=0.6462
p=0.6096
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

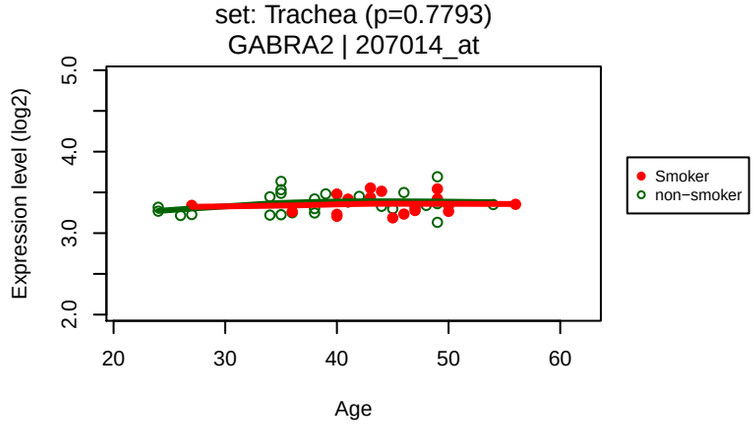

Gene expression ratio in Males
gene: GABRA2 | 207014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2806
p=0.6556
p=0.2549
p=0.9736
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA2 | 207014_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2647
p=0.398
p=0.4659
p=0.487
SmallAirway2
LargeAirway2
Trachea
Macrophage2

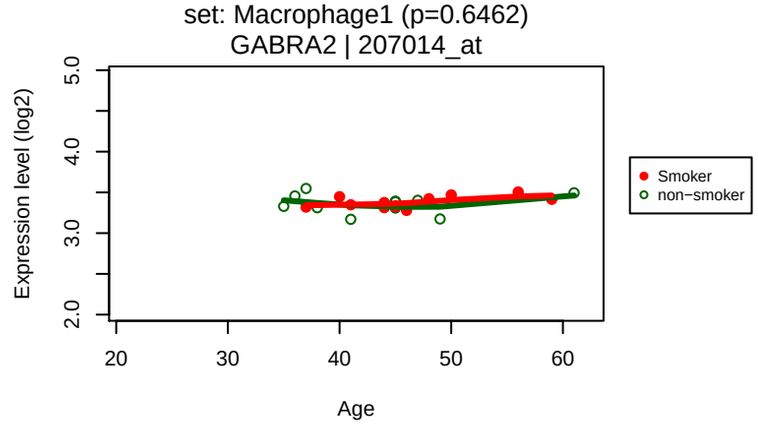

set: SmallAirway1 (p=0.5377)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker

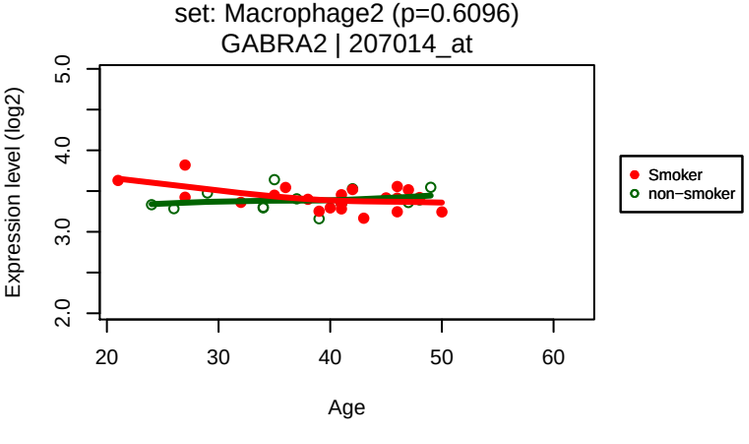

set: SmallAirway2 (p=0.1134)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker

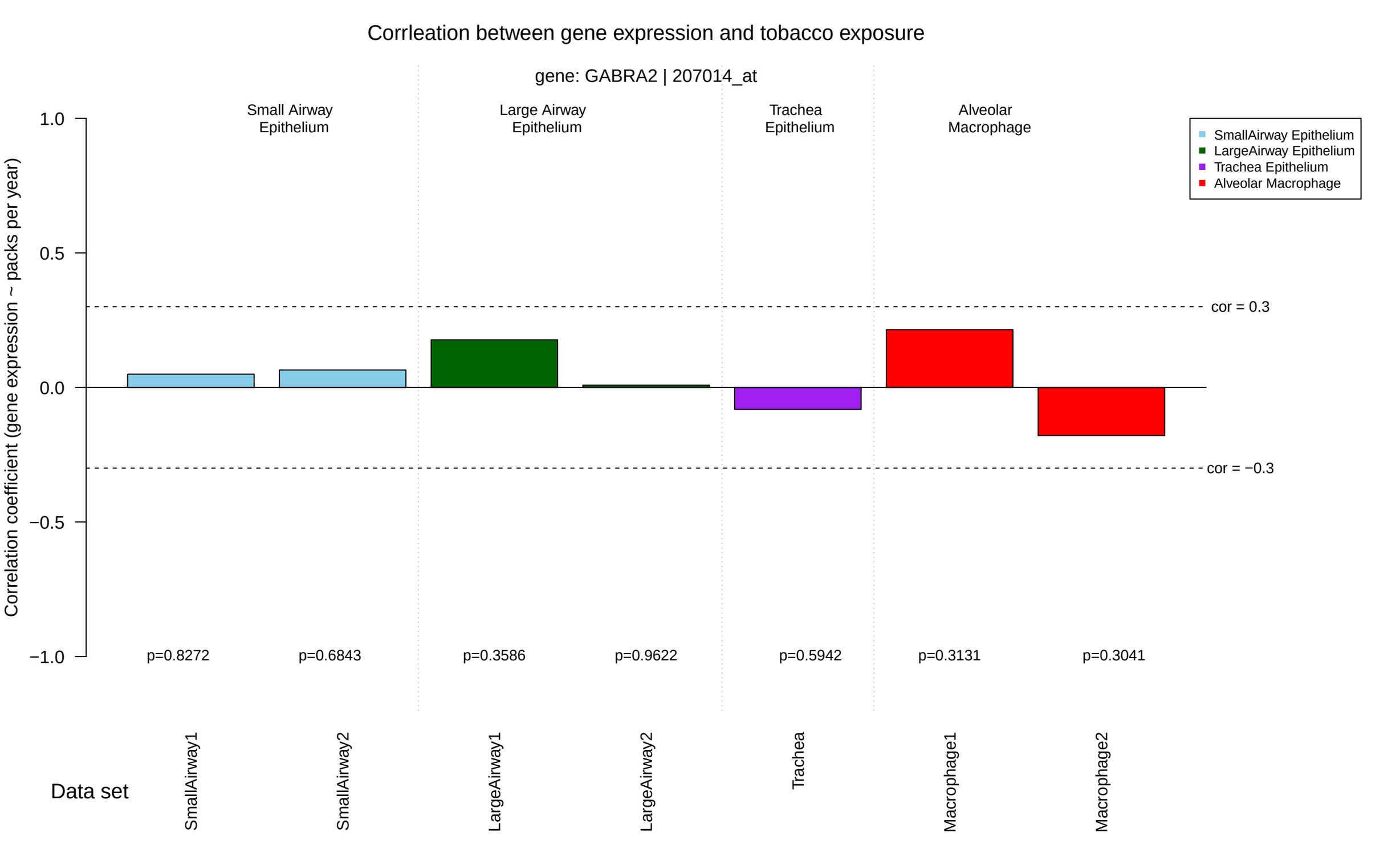

set: LargeAirway1 (p=0.3783)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7942)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7793)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6462)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6096)
GABRA2 | 207014_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA2 | 207014_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8272
p=0.6843
p=0.3586
p=0.9622
p=0.5942
p=0.3131
p=0.3041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2